Prabhavit Gupta

Software Engineer | Agent Platform, Headless Browser Automation (Playwright/CDP) & Developer Tools

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Executive Summary

- **Agent Platform & Developer Tooling:** Software engineer with 3 years commercial software experience specializing in autonomous agent execution harnesses, browser automation frameworks, and robust developer APIs across production deployments.
- **Browser Automation, Playwright & CDP Depth:** Deep hands-on expertise orchestrating headless Chromium instances, Chrome DevTools Protocol (CDP), Playwright test/scraping suites, Model Context Protocol (MCP) tool integrations, and handling DOM mutation reliability under complex web conditions.
- **High-Agency Execution & Full-Stack Delivery:** Fluent across TypeScript and Python, with a track record building resilient developer-facing services, stateful web session handlers, and automated evaluation harnesses in fast-paced engineering environments.

Technical Skills & Core Competencies

- **Browser Automation & Protocols:** Playwright, Puppeteer, Chrome DevTools Protocol (CDP), Headless Browser Infrastructure, DOM Traversal, Web Scraping (Playwright, BeautifulSoup), Anti-Bot Navigation, Session State Management.
- **Agent Systems & AI Tooling:** Model Context Protocol (MCP), Autonomous Agent Tool Loops, Tool Calling, Structured Outputs (Zod, Pydantic), Prompt Engineering, Agent Evaluation Suites (DeepEval).
- **Core Software Engineering:** TypeScript, Node.js, Python (AsyncIO, Typing, PyTest), REST/GraphQL APIs, WebSockets, Sandboxed Tool Execution, SQLite, PostgreSQL.
- **DevOps & Infrastructure:** Docker, GitHub Actions, CI/CD, Postman, Distributed Tracing (Langfuse), Linux CLI, Git.

Professional Experience

AI Software Engineer & Agent Systems Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote / Hybrid)

- Engineered agentic browser execution harnesses using Playwright and Model Context Protocol (MCP), connecting LLMs to live web pages with reliable DOM element interaction, scrolling, and action dispatch.
- Designed CDP-based automation clients inspecting DOM snapshots, network requests, and console events to evaluate agent execution fidelity and diagnose failure modes.
- Built custom developer tools and API connectors enabling autonomous agents to safely authenticate, extract structured data, and submit web forms with validation barriers.
- Implemented automated trace monitoring with Langfuse, tracking token utilization, browser interaction latency, and tool dispatch success across hundreds of test scenarios.
- Authored technical integration documentation, test harnesses, and reusable SDK examples for external developer teams.

Lead Developer & Product Engineer • *FlowSpace.co*
Mar 2024 – Aug 2026 • London, UK (Remote)

- Scaled platform from 2,000 to 4,500+ active users, designing core features, real-time platform integrations, and automated customer-facing agents.
- Migrated the live production platform off a legacy stack onto Next.js and Laravel without downtime or service disruption for members.
- Deployed autonomous customer support agents querying live booking databases with strict validation boundaries and human-in-the-loop escalation.
- Built automated data ingestion and web extraction pipelines aggregating unstructured venue data across multiple European markets.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Shipped automation services and API integrations in Python and TypeScript, connecting disparate enterprise systems and databases.
- Implemented continuous integration pipelines with automated test suites in Docker, reducing deployment cycle times.
- Collaborated directly with business end-users to understand operational workflows and translate requirements into reliable software tools.

Key Engineering Projects

- **Autonomous Browser Agent Runtime (Playwright + CDP + MCP + TypeScript):** Architected a sandboxed execution layer allowing AI agents to navigate web applications, inspect accessibility trees, execute deterministic actions, and extract structured data via CDP.
- **Automated Multi-Source Extraction Pipeline (Python + Playwright + Claude API):** Built a crawling engine processing 900+ data sources with dynamic JavaScript rendering, human review checkpoints, and automated schema enforcement.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy, International Studies & Economics** – Ca' Foscari University of Venice (2020 – 2024)
  Rigorous foundation in formal logic, epistemology, decision theory, and complex systems analysis.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Legal Status & Availability:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy). Open to international relocation to San Francisco with visa sponsorship (H-1B, O-1, etc.); available immediately.